

3.7 Debugging Code in Python

ڈیباگنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 24 of 24

3.7 Debugging Code in Python

LEARN IT

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly. Every program you write will have bugs. The skill is not "never make a mistake"; it is "find the mistake quickly".

Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>
Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

WORKED EXAMPLE

Reading a traceback. When Python crashes it prints a roadmap: the file, the line number, and the error type.

```
x = int("abc")
```

OUTPUT

```
Traceback (most recent call last):
  File "debug_program.py", line 1, in <module>
    x = int("abc")
ValueError: invalid literal for int() with base 10: 'abc'
```

Diagnosis: on **line 1** we tried to convert the letters "abc" into an integer, which raised a `ValueError`.

EXAM TIP

Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong answer? → logical. Write the name of the error **and** how to fix it.

REMEMBER IT

Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

COMMON MISTAKE

A logical error is the most dangerous because Python reports nothing. Always test with values whose answer you already know (for a rectangle 3 by 4, area must be 12, not 7).

اردو میں سمجھیں

ڈیبگنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔ نحوی غلطی پر پروگرام چلتا ہی نہیں، رن ٹائم پر چلتے چلتے رک جاتا ہے، منطقی غلطی پر چل تو جاتا ہے مگر جواب غلط دیتا ہے — یہ سب سے خطرناک ہے۔

CHECK YOURSELF

1. `print("Hi"` — which error type?
2. A program that should add two numbers multiplies them instead. Which error type?
3. `int("12.5")` raises which named exception?

Answers: 1. Syntax error. 2. Logical error. 3. `ValueError` (runtime).

Next lecture: Chapter review (in the chapter lecture PDF)